

RESEARCH INTERESTS

My research lies at the intersection of **Human-AI Interaction**, **Reinforcement Learning**, and **Natural Language Processing**. I study how large language models can acquire the linguistic and strategic nuance required for complex human interaction—settings such as negotiation and diplomacy, where success depends on implicit signals like trust, rapport, and long-horizon collaboration. My current work fine-tunes LLMs through **supervised** and **reinforcement learning** approaches (**SFT**, **REINFORCE**, **DPO**, **GRPO**) to align model behavior with these latent reward structures. The broader goal is to build AI that augments human *decision-making* in language-rich, socially complex environments.

EDUCATION

University of Maryland

Doctor of Philosophy (Ph.D.) in Computer Science

College Park, MD

Aug. 2021 – Present

- Advisor: [Prof. Jordan Boyd-Graber](#)
- Funded by: DARPA-SHADE (Diplomat AI), DARPA-FACT (Friction for humans using LLMs)
- Ph.D. Candidacy since Fall 2025
- Expected Graduation: Fall 2026

Master of Science (M.S.) in Computer Science

Aug. 2019 – May 2021

Chulalongkorn University

Bachelor of Science (B.S.) in Computer Science & Mathematics

Bangkok, Thailand

May 2013 – May 2017

EXPERIENCE

Graduate Research Assistant

University of Maryland

Jan. 2022 – Present

College Park, MD

- Supported by DARPA-FACT; PI: [Jonathan May \(USC\)](#) and co-PIs: [Jordan Boyd-Graber \(UMD\)](#) and [Jonathan K. Kummerfeld \(USydney\)](#).
- *In progress: RL optimization in multi-turn negotiation tasks:* LLMs struggle with long-horizon tasks and scarce rewards, we prompt-tuning (using DSPy) and fine-tuning Llama 3.1 using SFT, DPO and GRPO (RLVR with rubric rewards), to improve LLMs. Negotiation tasks require multi-aspect to achieve, we derive this problem from rubric rewards into weak-to-strong feedback, modifying GRPO.
- *Highlight projects:*
 - * [CTRL-D](#): grounding natural language space into action space, using counterfactual RL for value estimation to detect lies. 
 - * [AI is Less Cooperation](#): benchmarking Meta's Cicero to highlight its competence against humans, and building Cicero to assist beginners to make decisions and talk like experts. 

Software Engineering (Ph.D.) Intern

Google LLC

May – Aug. 2026

Mountain View, CA

- Developing graph-based advertiser embeddings that fuse user behavior, textual, and network signals to replace sparse feature representations.
- Applying Graph Neural Networks to improve personalization and prediction accuracy in advertising and recommendation systems.

Business Intelligence Developer

True Digital Group

Jan. 2019 – June 2019

Bangkok, Thailand

- Provided Tableau training to Business Intelligence and Analytics team
- Lead BI team with Agile methodology

Business Intelligence Developer

Agoda Inc.

June 2017 – June 2018

Bangkok, Thailand

- Delivered BI Solutions for PPC marketing
- ETL and Visualized PPC data using SQL, Tableau and Google Data Studio

PUBLICATIONS

- **W. Wongkamjan**, Y. Wang, F. Gu, D. Peskoff, J. K. Kummerfeld, J. May, J. Boyd-Graber. “Should I Trust You? Detecting Deception in Negotiations using Counterfactual RL.” *Findings of the Association for Computational Linguistics* (ACL Findings), 2025. [paper](#) [code](#)
- F. Gu, **W. Wongkamjan**, J. K. Kummerfeld, D. Peskoff, J. May, J. Boyd-Graber. “Personalized Help for Optimizing Low-Skilled Users’ Strategy.” *The Nations of the Americas Chapter of the Association for Computational Linguistics* (NAACL), 2025. [paper](#)
- **W. Wongkamjan**, F. Gu, Y. Wang, U. Hermjakob, J. May, B. Stewart, J. K. Kummerfeld, D. Peskoff, J. Boyd-Graber. “More Victories, Less Cooperation: Assessing Cicero’s DIPLOMACY Play.” *Annual Meeting of the Association for Computational Linguistics* (ACL), 2024. [paper](#)
- N. Nananukul, **W. Wongkamjan** “What if Red Can Talk? Dynamic Dialogue Generation Using Large Language Models.” *Wordplay: When Language Meets Games* (ACL Workshop), 2024. [paper](#)
- X. Wang, R. Zheng, Y. Sun, R. Jia, **W. Wongkamjan**, H. Xu, F. Huang. “COPlanner: Plan to Roll Out Conservatively but to Explore Optimistically for Model-Based RL.” *International Conference of Learning Representation* (ICLR), 2024. [paper](#)
- X. Wang, **W. Wongkamjan**, R. Jia, F. Huang. “Live in the Moment: Learning Dynamics Model Adapted to Evolving Policy.” *International Conference on Machine Learning* (ICML), 2023. [paper](#)

PROFESSIONAL SERVICE

- Reviewer: NeurIPS 2025, ARR 2024–2025, CHI 2025, AAAI GenPlan 2025.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, R, SQL, MATLAB

Deep Learning: PyTorch, TensorFlow

Data Systems: Hadoop, BigQuery

Data Tools: Tableau, Google Data Studio

Other: Unreal Engine 4